NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

1^{st} **year**
Discrete Mathematics I
2023/2024

Final Exam Solution

Exercice 1. *At a particular company, employees have a seven digit security code. Five of the digits must come from $\{1, 4, 5, 7\}$ with repetition allowed. The remaining digits must come from $\{2, 3, 9\}$ with no repetition.*

1. *How many security codes are possible?*
2. *How many of these begin with 1?*
3. *How many of these end with 5?*
4. *How many of these neither begin with 1 nor end with 5?*
5. *Suppose that this company has 150,000 employees. Should they be that two employees have the same security code?*

Solution:

1. First we have to choose the position for the five digit in $\binom{7}{5}$ ways then we have 4^5 ways to arrange the five digits with repetition is allowed, and 3×2 to arrange the remaining 2 digits such no repetition is allowed by (MP), the total number is $\binom{7}{5} \times 4^5 \times 3 \times 2 = 129024$. (0.75 pts)
2. we first fix 1 in the first position, then choose the position for the remaining four digit in $\binom{6}{4}$ ways then we have 4^4 ways to arrange the five digits with repetition is allowed, and 3×2 to arrange the remaining 2 such no repetition is allowed by (MP), the total number is $\binom{6}{4} \times 4^4 \times 3 \times 2 = 23040$. (0.5 pts)
3. the same as the previous question, we just fix 5 in the last position. $\binom{6}{4} \times 4^4 \times 3 \times 2 = 23040$. (0.25 pts)
4. we first find the number of security code such that the first number is 1 and the last is 5 and subtract from the total number, which is given by $\binom{7}{5} \times 4^5 \times 3 \times 2 - \binom{5}{3} \times 4^3 \times 3 \times 2 = 67584$. (0.75 pts)
5. Yes, it is possible that two employees have the same security code, because we have $\binom{7}{5} \times 4^5 \times 3 \times 2 = 129024 < 150000$. (0.25 pts)

Exercice 2. *I) Twenty different books are to be put on five book shelves, each of which holds at least twenty books.*

1. *How many different arrangements are there if you only care about the number of books on the shelves (and not which book is where)?*
2. *How many different arrangements are there if you care about which books are where, but the order of the books on the shelves doesn't matter?*
3. *How many different arrangements are there if the order on the shelves does matter?*

II)
1. *Let n and k be positive integers. Give a combinatorial proof that $\sum_{k=1}^{n} k\binom{n}{k}^2 = n\binom{2n-1}{n-1}$.*
2. *What is the coefficient of $x^{30}y^{20}z^{50}$ in the expansion of $(x - 2y + 3z - \frac{1}{2}w)^{100}$?*

Solution:

I) 1. The total number of different arrangement if we only care about the number of books on the shelves (and not which book is where) is the number of solution of the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 = 20,\ x_i \geq 0,\ i = 1, ..., 5 \tag{1}$$

The number of solution of the equation (1) is $\binom{20+5-1}{5-1} = \binom{24}{4} = 10626$. (1 *pts*)

2. Total number of arrangement if we you care about which books are where, but the order of the books on the shelves doesn't matter is an arrangement with repetition $\underbrace{5 \times 5 \times \cdots \times 5}_{20\ times} = 5^{20}$. (0.75 *pts*)

3. The total number of different arrangements if the order on the shelves does matter. we first find the number of ways to arrange the book on the shelves which is answered in the first question, then multiply by the number of permutations of the books representing their order, we get $\binom{24}{4} 20! = 10626 \times 20!$. (0.5 *pts*)

II) 1. We have a group of $2n$ people, n boys and n girls. We want to select a committee of n people, including a boy as a president.

We can select the president first. There are n ways to do this. For each of these ways, there are $\binom{2n-1}{n-1}$ ways to choose the rest of the committee. Thus by (MP) the total number is $n\binom{2n-1}{n-1}$ ways to choose a committee of n people with a boy as a president. On the other hand, we now count on the LHS the number of committees with a boy as a president in a different way. The committee will have a total of k boys, where $k = 1, \ldots, n$, and $n-k$ girls.

We can select k boys and $n-k$ girls in $\binom{n}{k}\binom{n}{n-k}$ ways. For each of these ways, the president can be chosen from the k boys in k ways. Thus the total number is $k\binom{n}{k}\binom{n}{n-k}$ ways to make a committee with k boys, including a boy as a president, and $n-k$ girls.

Finally, note that $\binom{n}{n-k} = \binom{n}{k}$, and sum over all k. The number of committees with boy president is $\sum_{k=1}^{n} k\binom{n}{k}^2 = n\binom{2n-1}{n-1}$. (1 *pts*)

2. the coefficient of $x^{30}y^{20}z^{50}$ in the expansion of $(x - 2y + 3z - \frac{1}{2}w)^{100}$ is

$$\binom{100}{30, 20, 50, 0} (-2)^{20}\, 3^{50}. \text{ (0.5 } pts\text{)}$$

Exercice 3. *A runner has four weeks before a one-kilometer race. He has decided to run a whole number of kilometers each day, at least one kilometer per day, and no more than 12 kilometers in any one week. Show that there exists a succession of days during which the runner will have run exactly 7 kilometers. Would the conclusion be still correct if we replace 7* *with 6? What about if we replaced 7 with 8?*

Solution: Let the list of integers be $x_1, x_2, \ldots, x_{28}$. Let $s_1, s_2, \ldots, s_{28}$ be the sequence of the partial sums. In other words,

$$\begin{aligned} s_1 &= x_1, \\ s_2 &= x_1 + x_2, \\ &\vdots \\ s_{28} &= x_1 + x_2 + \cdots + x_{28}. \ \textcolor{red}{(0.75\ pts)} \end{aligned}$$

Note that we know $s_{28} \leq 12 \times 4 = 48$, and $1 \leq s_1 < s_2 < \cdots < s_{28} \leq 48$. Now we just add 7 to each s_i to get a new integer t_i. In other words,

$$\begin{aligned} t_1 &= s_1 + 7 \\ t_2 &= s_2 + 7 \\ &\vdots \\ t_{28} &= s_{28} + 7. \ \textcolor{red}{(0.75\ pts)} \end{aligned}$$

Now, $t_{28} \leq 48 + 7 = 55$, and we have $8 \leq t_1 < t_2 < \cdots < t_{28} \leq 55$. Consider a list of 55 integers (sums) consisting of both the $s's$ and the $t's$:

$$1 \leq s_1, s_2, \ldots, s_{28}, t_1, t_2, \ldots, t_{28} \leq 55.$$

We have 56 integers, and each is an integer no smaller than 1 and no larger than 55. The pigeonhole principle says that two of them must be the same. The $s's$ can't be equal to each other and the $t's$ can't be equal to each other either. So, at least one of the $s's$ must be equal to one of the $t's$. (0.5 pts) Hence, we have

$$s_j = t_i = s_i + 7,$$

for some $1 \leq i < j \leq 28$. This means that $s_j - s_i = 7$, but

$$\begin{aligned} 7 = s_j - s_i &= (x_1 + \cdots + x_j) - (x_1 + \cdots + x_i) \\ &= x_{i+1} + x_{i+2} + \cdots + x_j. \end{aligned}$$

We conclude that the sum of the $(i+1)$th integer all the way to the j th integer is exactly 7. (0.5 pts)
If we replace 7 with 6: The same argument holds, and there will exist a succession of days during which the runner will have run exactly 6 miles. ; (0.25 pts)
but if we replace 7 by 8, the argument does not hold, because we will have as much sums as possible values, therefore we can not conclude the result. (0.5 pts)

Exercice 4. *Let $n \in \mathbb{N}$, discuss the number of integer solutions in $\mathbb{N} \cup \{0\}$ of the following inequality*

$$x_1 + x_2 + x_3 + x_4 + x_5 \leq n,$$

with $1 \leq x_1 \leq 3,\ 3 \leq x_2 \leq 7,\ 4 \leq x_3 \leq 6, 2 \leq x_4 \leq 4,$ *and* $x_5 = 1$.

Solution: The number of solution in $\mathbb{N} \cup \{0\}$ of the inequality

$$x_1 + x_2 + x_3 + x_4 + x_5 \leq n$$

with $1 \leq x_1 \leq 3,\ 3 \leq x_2 \leq 7,\ 4 \leq x_3 \leq 6,\ 2 \leq x_4 \leq 4$, and $x_5 = 1$. equivalent to find the number of solution of the equation,

$$x_1 + x_2 + x_3 + x_4 = m - 1 \qquad \text{for } m = 11, \ldots, n$$

with $1 \leq x_1 \leq 3,\ 3 \leq x_2 \leq 7,\ 4 \leq x_3 \leq 6,\ 2 \leq x_4 \leq 4$. $(0.25\ pts)$
We set the following change of variables $x_1' = x_1 - 1,\ x_2' = x_2 - 3,\ x_3' = x_3 - 4,\ \ x_4' = x_4 - 2$ we get $x_1' + x_2' + x_3' + x_4' = m - 11$ with $0 \leq x_1' \leq 2, 0 \leq x_2' \leq 4, 0 \leq x_3' \leq 2, 0 \leq x_4' \leq 2$. $(0.75\ pts)$
There are no solutions for $n \leq 10$.
The possible values of n are $11 \leq n \leq 21$.
We have $m = 11, \ldots, n$ and $\Omega_m = \{(x_1', x_2', x_3', x_4') : x_1' + x_2' + x_3' + x_4' = m - 11\}$,
$|\Omega_m| = \binom{m - 11 + 4 - 1}{4 - 1} = \binom{m-8}{3}$. $(0.5\ pts)$
Let
$A_1 = \{(x_1', x_2', x_3', x_4') : x_1' + x_2' + x_3' + x_4' = m - 11,\ x_1' \geqslant 3\}$,
$A_2 = \{(x_1', x_2', x_3', x_4') : x_1' + x_2' + x_3' + x_4' = m - 11,\ x_2' \geqslant 5\}$,
$A_3 = \{(x_1', x_2', x_3', x_4') : x_1' + x_2' + x_3' + x_4' = m - 11,\ x_3' \geqslant 3\}$,
$A_4 = \{(x_1', x_2', x_3', x_4') : x_1' + x_2' + x_3' + x_4' = m - 11,\ x_4' \geqslant 3\}$,
$(0.5\ pts)$

$|\overline{A_1}\ \overline{A_2}\ \overline{A_3}\ \overline{A_4}| = |\Omega_m| - |A_1 \cup A_2 \cup A_3 \cup A_4|$
$= |\Omega_m| - |A_1| - |A_2| - |A_3| - |A_4| + |A_1A_2| + |A_1A_3| + |A_1A_4| + |A_2A_3| + |A_2A_4| + |A_3A_4| - |A_1A_2A_3| - |A_1A_2A_4| - |A_1A_3A_4| - |A_2A_3A_4| + |A_1A_2A_3A_4|$ $(0.25\ pts)$

$$|A_1| = \binom{m-11}{3},\ |A_2| = \binom{m-13}{3},\ |A_3| = \binom{m-11}{3},\ |A_4| = \binom{m-11}{3}.$$

$$|A_1A_2| = \binom{m-16}{3},\ |A_1A_3| = \binom{m-14}{3},\ |A_1A_4| = \binom{m-14}{3},\ |A_2A_3| = \binom{m-16}{3},$$
$$|A_2A_4| = \binom{m-16}{3},\ |A_3A_4| = \binom{m-14}{3}.$$

$$|A_1A_2A_3| = \binom{m-19}{3} = 0, |A_1A_3A_4| = \binom{m-17}{3}, |A_1A_2A_4| = \binom{m-19}{3} = 0, |A_2A_3A_4| = \binom{m-19}{3} = 0.$$

$$|A_1A_2A_3A_4| = \binom{m-22}{3} = 0. \ (1\ pts)$$

Thus the number of solution is

$$\sigma_n = \sum_{m=11}^{n} \left[\binom{m-8}{3} - 3\binom{m-11}{3} - \binom{m-13}{3} + 3\binom{m-14}{3} + 3\binom{m-16}{3} - \binom{m-17}{3}. \right] (0.5\ pts)$$

- **2^{nd} approach** :

The number of solution in $\mathbb{N} \cup \{0\}$ of the inequality

$$x_1 + x_2 + x_3 + x_4 + x_5 \leq n$$

with $1 \leq x_1 \leq 3,\ 3 \leq x_2 \leq 7,\ 4 \leq x_3 \leq 6,\ 2 \leq x_4 \leq 4,$ and $x_5 = 1$. equivalent to find the number of solution of the equation,

$$x_1 + x_2 + x_3 + x_4 + x_6 = n - 1$$

with $1 \leq x_1 \leq 3,\ 3 \leq x_2 \leq 7,\ 4 \leq x_3 \leq 6,\ 2 \leq x_4 \leq 4,\ x_6 \geq 0$. (0.25 *pts*)
We set the following change of variables $x_1' = x_1 - 1,\ x_2' = x_2 - 3,\ x_3' = x_3 - 4,\ \ x_4' = x_4 - 2,\ x_5' = x_6$ we get
$x_1' + x_2' + x_3' + x_4' + x_5' = n - 11$ with $0 \leq x_1' \leq 2, 0 \leq x_2' \leq 4, 0 \leq x_3' \leq 2, 0 \leq x_4' \leq 2, x_5' \geq 0$ (0.75 *pts*)
There are no solutions for $n \leq 10$.
The possible values of n are $11 \leq n \leq 21$.
$\Omega_n = \{(x_1', x_2', x_3', x_4', x_5') : x_1' + x_2' + x_3' + x_4' + x_5' = n - 11\}$,
$|\Omega_m| = \binom{n-11+5-1}{5-1} = \binom{n-7}{4}$. (0.5 *pts*)
Let
$A_1 = \{(x_1', x_2', x_3', x_4', x_5') x_1' + x_2' + x_3' + x_4' + x_5' = n - 11,\ x_1' \geqslant 3\}$,
$A_2 = \{(x_1', x_2', x_3', x_4', x_5') x_1' + x_2' + x_3' + x_4' + x_5' = n - 11,\ x_2' \geqslant 5\}$,
$A_3 = \{(x_1', x_2', x_3', x_4', x_5') x_1' + x_2' + x_3' + x_4' + x_5' = n - 11,\ x_3' \geqslant 3\}$,
$A_4 = \{(x_1', x_2', x_3', x_4', x_5') x_1' + x_2' + x_3' + x_4' + x_5' = n - 11,\ x_4' \geqslant 3\}$,
(0.5 *pts*)

$|\overline{A_1}\ \overline{A_2}\ \overline{A_3}\ \overline{A_4}| = |\Omega_n| - |A_1 \cup A_2 \cup A_3 \cup A_4|$
$= |\Omega_n| - |A_1| - |A_2| - |A_3| - |A_4| + |A_1A_2| + |A_1A_3| + |A_1A_4| + |A_2A_3| + |A_2A_4| + |A_3A_4| - |A_1A_2A_3| - |A_1A_2A_4| - |A_1A_3A_4| - |A_2A_3A_4| + |A_1A_2A_3A_4|$ (0.25 *pts*)

$$|A_1| = \binom{n-10}{4},\ |A_2| = \binom{n-12}{4},\ |A_3| = \binom{n-10}{4},\ |A_4| = \binom{n-10}{4}.$$

$$|A_1A_2| = \binom{n-15}{4},\ |A_1A_3| = \binom{n-13}{4},\ |A_1A_4| = \binom{n-13}{4},\ |A_2A_3| = \binom{n-15}{4},$$
$$|A_2A_4| = \binom{n-15}{4},\ |A_3A_4| = \binom{n-13}{4}.$$

$$|A_1A_2A_3| = \binom{n-18}{4} = 0, |A_1A_3A_4| = \binom{n-16}{4}, |A_1A_2A_4| = \binom{n-18}{4} = 0, |A_2A_3A_4| = \binom{n-18}{4} = 0.$$

$$|A_1A_2A_3A_4| = \binom{n-21}{4} = 0. \ (1\ pts)$$

Thus the number of solution is

$$\sum_{n=11}^{21}\left[\binom{n-7}{4} - 3\binom{n-10}{4} - \binom{n-12}{4} + 3\binom{n-13}{4} + 3\binom{n-15}{4} - \binom{n-16}{4}.\right] (0.75\ pts)$$

Exercice 5. *For all $n, m \in \mathbb{N}$, let $L(n,m)$ be the number of ways to distribute n distinguishable balls into m indistinguishable boxes such that no box is empty, using the inclusion-exclusion principle, find a formula for $L(n,m)$.*

Solution: Let Ω denote the set of all distributions of n distinguishable balls into m distinguishable boxes. Note that $|\Omega| = m^n$. $(0.5\ pts)$ Let A_i denote the set of all distributions in which the box i is empty $(0.25\ pts)$. Thus by the Principle of Inclusion-Exclusion, the number of distributions is given by

A_1 is the set of all distribution of n distinguishable balls where the first box is empty. $|A_1| = (m-1)^n$, thus $|A_i| = (m-1)^n$ for each $i = 1, 2, 3, \dots, m$, Hence,

$$\sum_{i=1}^{m} |A_i| = \binom{m}{1}(m-1)^n. \ (0.5\ pts)$$

Similarly, let A_1A_2 be the set of all distribution of n distinguishable balls where the first and the second boxes are empty, so, we have $|A_1A_2| = (m-2)^n$. Thus

$$|A_iA_j| = (m-2)^n, \qquad \sum_{1\le i<j\le m} |A_iA_j| = \binom{m}{2}(m-2)^n. \ (0.5\ pts)$$

Using the same arguemnt $|A_{i_1}A_{i_2}\cdots A_{i_k}|$ for $k = 1, 2, \dots, m$. We have $|A_{i_1}A_{i_2}\cdots A_{i_m}| = (m-k)^n$. This implies that $\displaystyle\sum_{1\le i_1<i_2<\ldots<i_k\le m} |A_{i_1}A_{i_2}\cdots A_{i_k}| = \binom{m}{k}(m-k)^n$. $(0.5\ pts)$

Hence, the number of distribution of n distinguishable balls into m distinguishable boxes such that no box is empty is equal to

$$\begin{aligned}
|\overline{A_1}\,\overline{A_2}\cdots\overline{A_m}| &= |\Omega| - |A_1 \cup A_2 \cdots \cup A_m| \\
&= m^n - \sum_{k=1}^{m}(-1)^{k-1} \sum_{1\le i_1<i_2<\cdots<i_k\le m} |A_{i_1}\cap A_{i_2}\cap\cdots\cap A_{i_k}| \\
&= m^n - \sum_{k=1}^{m}(-1)^{k-1}\binom{m}{k}(m-k)^n \\
&= \sum_{k=0}^{m}(-1)^k\binom{m}{k}(m-k).^n (1\ pts)
\end{aligned}$$

Now, it remains to remove the order of the boxes by dividing by the number of their permutations

$$k!L(n,m) = \sum_{k=0}^{m}(-1)^k\binom{m}{k}(m-k)^n$$
$$L(n,m) = \frac{1}{k!}\sum_{k=0}^{m}(-1)^k\binom{m}{k}(m-k)^n. \ (0.75\ pts)$$

Exercice 6. *I) Let G be a graph with $V(G) = 1, 2, \ldots, 10$, such that two numbers i and j in $V(G)$ are adjacent if and only if $i + j$ is a multiple of 4.*

(a) Draw G and determine $e(G)$.

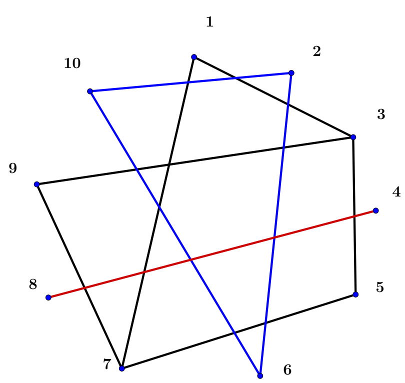
(b) Is G connected?

(c) How many components does G have?

(d) Determine the edge set for the subgraph of G induced by $W = \{1, 3, 7, 9\}$.

II) Prove that if for a graph G of order 9 every pair of distinct vertices $u, v \in V(G)$ where $deg(u) + deg(v) \geqslant 8$ then G is connected.

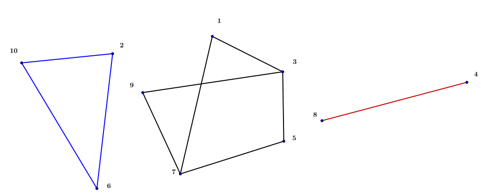
Solution:



I) a) $e(G) = 10$. $(0.5\ pts)$

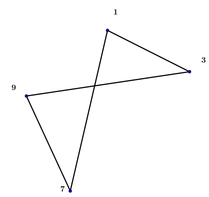
b) No, G is diconnected, because if we take the vertices 4 and 9 they are not connected. $(0.25\ pts)$

c) G has 3 components. $(0.5\ pts)$



d) The edge set induced by W is $\{13, 17, 37, 79\}$. $(0.5\ pts)$

Final Exam Solution



II) Suppose for a contradiction that G is a disconnected graph of order 9 such that for every pair of distinct vertices u and v we have $deg(u) + deg(v) \geqslant 8$.
Since G is disconnected there must exist vertices, say x and y, such that no xy path exists in G. If no xy path exists then certainly $xy \notin E(G)$ and x, y do not share neighbours. Let $deg(x) = k$ for some $k = 0, ..., 8$. Since y is not adjacent to x and they share no neighbours, $deg(y) \leqslant 8 - k - 1$.
Together we have $deg(x) + deg(y) \leqslant k + 8 - k - 1 = 7$, a contradiction. $(0.75\ pts)$
Therefore our additional hypothesis that G was disconnected was false, hence G is connected.

Exercice 7 (Optional). *Let n and m be two positive integers such that $n = km + r$ for some $k \in \mathbb{Z}$ and $r \in \{0, 1, 2, ..., m - 1\}$.*

1. *Determine the number of integer solutions to the equation:*
$$x_1 + x_2 + \cdots + x_n + y_1 + \cdots + y_m = gcd(m, r)$$
subject to the constraints $x_i \geq \ell$ and $y_j \geq s$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$.
2. *How many solutions are there if m and n are coprime?*

Solution:

1. Using the known property that $gcd(n, m) = gcd(m, r) = an + bm$ for certain $a, b \in \mathbb{Z}$, the equation takes the form:
$$x_1 + x_2 + \cdots + x_n + y_1 + \cdots + y_m = an + bm$$
Subject to the conditions $x_i \geq \ell$ and $y_j \geq s$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$. By introducing new variables x_i and y_j constrained to be greater than or equal to 0, we obtain:
$$x_1 + x_2 + \cdots + x_n + y_1 + \cdots + y_m = n(a - \ell) + m(b - s)$$
With the additional constraints $x_i \geq 0$ and $y_j \geq 0$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$. The count of solutions is given by:
$$\left(\!\!\binom{n+m}{n(a-\ell)+m(b-s)}\!\!\right) = \binom{n(a-\ell+1)+m(b-s+1)-1}{n(a-\ell)+m(b-s)} = \binom{n(a-\ell+1)+m(b-s+1)-1}{n+m-1}. \ (1\ pts)$$
2. In the case where m and n are coprime (i.e., $gcd(m, n) = 1$), it is evident that selecting one variable as 1 and the rest as 0 yields solutions, then the count of solutions is given by:
   - $n + m$ solutions if $\ell = s = 0$.
   - 0 solutions otherwise. $(0.5\ pts)$